

DIPOLE TYPE BY LENGTH

l = antenna rod length (m)

λ (wavelength) = $\frac{c}{f} = \frac{3 \times 10^8}{f}$ (m)		
<i>l</i> /λ	Type	Use formula
< 1/50	Hertzian (short)	HERTZIAN $S(R, \theta)$, $D=1.5$
= 1/4	Quarter-wave	ARB formula, $\pi l/\lambda = \pi/4$
= 1/2	Half-wave	half-wave shortcut, $D=1.64$, $R_{\text{rad}}=73\ \Omega$
= 1	Full-wave	ARB, $D=2.41$, $R_{\text{rad}}=199\ \Omega$
other	Arbitrary	ARB formula

HERTZIAN DIPOLE — $S(R, \theta)$

Far-field E,H phasors small $l \ll \lambda$ $\vec{E}_\theta = \frac{jI_0 k l \eta_0}{4\pi R} e^{-jkR} \sin \theta$, $\vec{H}_\phi = \vec{E}_\theta / \eta_0$ Power density formulas $l/\lambda < 1/50$; far field	
Power density	$S(R, \theta) = \frac{S_{\text{max}} \sin^2 \theta}{\eta_0 k^2 I_0^2 l^2}$ (W/m ²)
Max pwr density	$S_{\text{max}} = \frac{32\pi^2 R^2}{32\pi^2 R^2}$ (W/m ²)
<i>l</i> : rod length (m)	I_0 : peak current (A)
<i>R</i> : obs. distance (m)	θ : angle from dipole axis (rad)
$k=2\pi/\lambda$ (rad/m)	$\eta_0=120\pi \approx 377\ \Omega$
Total radiated power	$P_{\text{rad}} = \frac{\eta_0 \pi}{3} (I_0 l/\lambda)^2$ (W)

ARBITRARY-LENGTH DIPOLE — $S(\theta)$

Power density at <i>R</i> any l
$S(\theta) = \frac{15 I_0^2}{\pi R^2} \left[\frac{\cos\left(\frac{\pi l}{\lambda} \cos \theta\right) - \cos\left(\frac{\pi l}{\lambda}\right)}{\sin \theta} \right]^2$
<i>At $\theta = 0, \pi$: bracket is 0/0. L'Hôpital $\Rightarrow S = 0$ (no axial radiation). TI-36X alt: evaluate bracket at $\theta = 0.001$ rad $\rightarrow$ tiny $\rightarrow 0$.</i>
Normalized pattern dimensionless $F(\theta) = S(\theta)/S_{\text{max}}$

COMMON ANGLES (deg $\leftrightarrow$ rad, trig values)

Conversion multiply by $\pi/180$ or $180/\pi$						
$\theta_{\text{rad}} = \theta_{\text{deg}} \cdot \frac{\pi}{180}$		$\theta_{\text{deg}} = \theta_{\text{rad}} \cdot \frac{180}{\pi}$				
°	rad	sin θ	cos θ	sin ² θ	cos ² θ	θ
0	0	0	1	0	1	
30	$\pi/6$	1/2	$\sqrt{3}/2$	1/4	3/4	
45	$\pi/4$	$\sqrt{2}/2$	$\sqrt{2}/2$	1/2	1/2	
60	$\pi/3$	$\sqrt{3}/2$	1/2	3/4	1/4	
90	$\pi/2$	1	0	1	0	
180	π	0	−1	0	1	

Antenna formulas usually take θ in rad. RAD mode on calc.

LINEAR ANTENNA ARRAY

Array factor (uniform, broadside) $k = 2\pi/\lambda$, $\gamma = kd \cos \theta$
$F_a(\theta) = \frac{\sin^2(N\gamma/2)}{\sin^2(\gamma/2)}$; $F_a^{\text{max}} = N^2$ at $\theta = \pi/2$
Normalize $F_a^{\text{norm}} = F_a/N^2$
HPBW (broadside, uniform) use Nd, not $(N-1)d$ $\beta \approx 0.88 \frac{\lambda}{Nd}$ (rad) = $50.8^\circ \cdot \frac{\lambda}{Nd}$
Electronic steering (scan to θ_0) replace γ with $\gamma' = kd(\cos \theta - \cos \theta_0)$ $\delta = \frac{2\pi d}{\lambda} \cos \theta_0$ ($\frac{\text{rad}}{\text{elt}}$); $\theta_0 = \frac{\pi}{2}$:broadside, 0:end-fire
Frequency scan feed-line incr. $\Delta \ell = n_0 \lambda_0$ $\cos \theta_0 \approx \frac{n_0 \lambda_0}{d} \frac{\Delta f}{f_0}$
<i>Grating lobes appear if $d > \lambda$. Avoid.</i>

BEAM PARAMETERS — β , Ω_p , *D*

Normalize always start here $F(\theta) = S(\theta)/S_{\text{max}}$ (unitless)		
Beamwidth β at threshold <i>X</i> Set $F(\theta_X) = X$, solve for θ_X ; $\beta = 2\theta_X$ (symmetric)		
Threshold	$X = 10^{\text{dB}/10}$	θ_X
HPBW (−3 dB)	0.5	$\sqrt{\frac{\ln 2}{a}}$
−6 dB	0.25	$\sqrt{\frac{\ln 4}{a}}$
−10 dB	0.1	$\sqrt{\frac{\ln 10}{a}}$
any $F(\theta_X)$	any <i>X</i>	$F(\theta_X) = X$
Pattern solid angle Ω_p recognize pattern, look up $\Omega_p = \int_0^{2\pi} \int_0^\pi F(\theta) \sin \theta \, d\theta \, d\phi$ (sr)		
<i>No ϕ dep.: $\Omega_p = 2\pi \int_0^\pi F(\theta) \sin \theta \, d\theta$. θ is the integration variable — DON'T plug in a number for θ (e.g. $\theta_{1/2}$). Integrate over $\theta \in [0, \pi]$.</i>		
Pattern $F(\theta)$	Ω_p (sr)	
Isotropic ($F=1$)	$4\pi \approx 12.57$	
Hertzian sin ² θ	$8\pi/3 \approx 8.38$	
Half-wave dipole	≈ 7.66	
Narrow Gauss $e^{-a\theta^2}$	$\pi/a = \pi\theta^2/2 / \ln 2$	
2-axis HPBW (aperture)	$\beta_{xz} \cdot \beta_{yz}$	
Other (use integral)	numerical	

TI-36X: [2nd] [d/dx] for $\int \square$, RAD mode, multiply by 2 π .

Directivity D <i>always $D = 4\pi/\Omega_p$; common shortcuts below</i>	
$D = \frac{4\pi}{\Omega_p}$ (unitless)	$D_{\text{dB}} = 10 \log_{10} D$
WORD TRAP: “direction” = θ_{max} (rad); “directivity” = D (unitless).	
Gain (if asked) ξ = radiation efficiency (unitless), $0 < \xi \leq 1$	$G = \xi D$ (unitless)
	$G_{\text{dB}} = 10 \log_{10} G$ (dB)
Lossless / ideal antenna $\Rightarrow G = D$.	

COMMON DIPOLE PARAMETERS

Lossless ($\xi=1$): $G = D$.				
Type	<i>l</i>	<i>D</i>	<i>D</i> _{dB}	<i>R</i> _{rad} (Ω)
Isotropic	N/A	1	0	—
Hertzian	$< \lambda/50$	1.5	1.76	$80\pi^2(l/\lambda)^2$
Half-wave	$\lambda/2$	1.64	2.15	73.2
Monopole	$\lambda/4$ (gnd)	1.64	2.15	36.6
Full-wave	λ	2.41	3.82	199

Half-wave: $P_{\text{rad}} = 36.6 I_0^2$ W. Monopole ($\lambda/4$ above ground): same patt. as half-wave but P_{rad} , R_{rad} halved.

ANTENNA AREAS (*A_e*, *A_p*)

Effective area antenna's RX cross section; <i>D</i> from BEAM PARAMS or table above
$A_e = \frac{\lambda^2 D}{4\pi}$ (m ²)
Physical cross section wire dipole, diameter <i>d</i>; <i>l</i> see table above
$A_p = l d$ (m ² , any dipole)
Inverse: <i>D</i> from <i>A_e</i>
$D = \frac{4\pi A_e}{\lambda^2}$ (unitless)
Thin wire: $A_e \gg A_p$ (e.g., Hertzian $A_e = 3\lambda^2/8\pi \approx 0.119\lambda^2$).

FRISIS TRANSMISSION FORMULA

For each antenna's *G*: if given by NAME (“half-wave dipole” etc.) $\rightarrow$ COMMON DIPOLE PARAMS ($G = D$, lossless). Else (dB/linear value given) $\rightarrow$ use as given (convert dB $\rightarrow$ linear).

Pwr density at RX, then RX pwr G_t, G_r linear gains; $\lambda = c/f$

$$S_r = \frac{G_t P_t}{4\pi R^2} \text{ (W/m}^2\text{); } \left[P_r = G_t G_r \left(\frac{\lambda}{4\pi R} \right)^2 P_t \right] \text{ (W)}$$

Equivalent: $P_r = S_r A_{e,r}$.

Solve for *R* max range

$$R = \frac{\lambda}{4\pi} \sqrt{\frac{G_t G_r P_t}{P_r}}$$

Equivalent form using *A_e* recv. area form

$$P_r = \frac{G_t P_t A_{e,r}}{4\pi R^2}, \quad G_r = \frac{4\pi A_{e,r}}{\lambda^2}$$

Free-space path loss: $L_{fs} = (4\pi R/\lambda)^2$, so $P_r = G_t G_r P_t / L_{fs}$.

Off-axis (patterns not aligned): multiply by $F_t(\theta_t, \phi_t) F_r(\theta_r, \phi_r)$, both = 1 when aligned.

Recipe 1. $\lambda = c/f$. 2. Convert any dB gain $\rightarrow$ linear: $G = 10^{G_{\text{dB}}/10}$.

3. Get G_t, G_r by antenna type: dipole $\rightarrow$ COMMON DIPOLE PARAMS; aperture $\rightarrow$ APERTURE box; arbitrary *F* $\rightarrow$ BEAM PARAMS. 4. Plug into Friis.

Use linear *G*, not dB. Convert FIRST.

dB $\leftrightarrow$ LINEAR

Works for any power-like quantity (*G*, *D*, *P_r*/*P_t*, SNR, ...):

$$X_{\text{dB}} = 10 \log_{10} X_{\text{lin}} \quad X_{\text{lin}} = 10^{X_{\text{dB}}/10}$$

Memorize: 3 dB $\Leftrightarrow \times 2$, 10 dB $\Leftrightarrow \times 10$. Add dB = multiply linear.

For E-field / amplitude use 20 log not 10 log (power $\propto E^2$).

NOISE & SNR

Noise power thermal noise into matched receiver

$$P_N = k_B T_{\text{sys}} B \text{ (W)}$$

$k_B = 1.38 \times 10^{-23}$ J/K; T_{sys} system noise temp (K); *B* receiver bandwidth (Hz).

Signal-to-noise ratio

$$\text{SNR} = \frac{P_{\text{rec}}}{P_N} \text{ (unitless)} \quad \text{SNR}_{\text{dB}} = 10 \log_{10} \frac{P_{\text{rec}}}{P_N}$$

APERTURE ANTENNAS (horn, dish)

A_p physical aperture area (m²); *A_e* effective area (m², = $\lambda^2 D/4\pi$); β_{xz}, β_{yz} HPBW's in two planes (rad); ℓ , *d* aperture side / diameter (m).

Directivity (any shape, uniform illum.)

$$D \approx \frac{4\pi A_p}{\lambda^2} \text{ (unitless)} \Rightarrow A_e \approx A_p$$

Directivity from beamwidths any aperture

$$D \approx \frac{4\pi}{\beta_{xz} \beta_{yz}} \text{ (use rad); circular: } D \approx 4\pi/\beta^2$$

HPBW and *D* by aperture shape uniform illum.; β in rad

Shape	<i>A_p</i>	HPBW (rad)	<i>D</i> (unitless)
Rect. $\ell_x \times \ell_y$	$\ell_x \ell_y$	$\beta_x \approx 0.88\lambda/\ell_x$ $\beta_y \approx 0.88\lambda/\ell_y$	$D \approx 4\pi/(\beta_x \beta_y)$ $= 4\pi \ell_x \ell_y / \lambda^2$
Square ℓ	ℓ^2	$\beta \approx 0.88\lambda/\ell$	$D \approx 4\pi/\beta^2 = 4\pi \ell^2 / \lambda^2$
Circular (diam. <i>d</i>)	$\pi d^2/4$	$\beta \approx 1.02\lambda/d$	$D \approx 4\pi/\beta^2 = \pi^2 d^2 / \lambda^2$

Scaling (any ratio) $D \propto A_p/\lambda^2$; $\beta \propto \lambda/\sqrt{A_p}$

$$D_{\text{new}} = D_{\text{old}} \cdot \frac{A_{p,\text{new}}}{A_{p,\text{old}}} \cdot \left(\frac{f_{\text{new}}}{f_{\text{old}}} \right)^2$$

$$\beta_{\text{new}} = \beta_{\text{old}} \cdot \sqrt{\frac{A_{p,\text{old}}}{A_{p,\text{new}}}} \cdot \frac{f_{\text{old}}}{f_{\text{new}}}$$

If a parameter is unchanged, its ratio = 1. Examples: double area $\rightarrow D \times 2$, $\beta \times 1/\sqrt{2}$; double freq $\rightarrow D \times 4$, $\beta \times 1/2$.

DIPOLE

<i>l</i> — antenna length (m)
$\lambda = c/f$ (m); $c = 3 \times 10^8$ m/s
I_0 — peak current (A)
$k = 2\pi/\lambda$ (rad/m)
<i>R</i> — obs. dist. (m); θ from axis
$\eta_0 = 120\pi \approx 377\ \Omega$
$S(R, \theta)$ — pwr density (W/m ²)

BEAM

$F(\theta) = S/S_{\text{max}}$ (unitless)
<i>a</i> — coef. on θ^2 in Gaussian
β — HPBW (rad or °)
Ω_p — pattern solid angle (sr)
<i>D</i> = $4\pi/\Omega_p$ — directivity
ξ radiation eff. (unitless); $G = \xi D$
$D_{\text{dB}} = 10 \log_{10} D$; $D = 10^{D_{\text{dB}}/10}$

FRISIS / dB

<i>P_t</i> — power transmitted (W)
<i>P_r</i> — power received (W)
<i>G_t</i> — transmit gain (linear)
<i>G_r</i> — receive gain (linear)
$S_r = G_t P_t / (4\pi R^2)$ (W/m ²)
$P_r = G_t G_r (\lambda/4\pi R)^2 P_t$
$G = 10^{G_{\text{dB}}/10}$
3 dB = $\times 2$, 10 dB = $\times 10$
$P_N = k_B T_{\text{sys}} B$ (W)

APERTURE

<i>A_p</i> — physical area (m ²)
$A_e = \lambda^2 D / (4\pi)$ (m ²)
$A_e \approx A_p$ for aperture
$D \approx 4\pi A_p / \lambda^2$ (unitless)
$\beta \approx 0.88\lambda/\ell$ (rect/sq.)
$\beta \approx 1.02\lambda/d$ (circular)
$2A_p \Rightarrow 2D$, $\beta/\sqrt{2}$

ARRAY

<i>N</i> — number of elements
<i>d</i> — spacing (m)
$\gamma = kd \cos \theta$
$F_a = \sin^2(N\gamma/2) / \sin^2(\gamma/2)$
$D \approx 4\pi A_p / \lambda^2$ (unitless)
$\beta \approx 0.88\lambda/d$ (circular)
$F_a^{\text{norm}} = F_a/N^2$
$\beta \approx 0.88\lambda/(Nd)$ (broadside)